A Particle Swarm Optimization Approach Based on Monte Carlo Simulation for Solving the Complex Network Reliability Problem

Wei-Chang Yeh, Yi-Cheng Lin, Yuk Ying Chung, and Mingchang Chih

***Abstract*—Reliability optimization has been a popular area of research, and received significant attention due to the critical importance of reliability in various kinds of systems. Most network reliability optimization problems are only focused on solving simple structured networks (e.g., series-parallel networks) of which the reliability function can be easily obtained in advance. However, modern networks are usually very complex, and it is impossible to calculate the exact network reliability function by using traditional analytical methods in limited time. Hence, a new Particle Swarm Optimization (PSO) based on Monte Carlo Simulation (MCS), named MCS-PSO, has been proposed to solve complex network reliability optimization problems. The proposed MCS-PSO can minimize cost under reliability constraints. To the best of our knowledge, this is the first attempt to use PSO combined with MCS to solve complex network reliability problems without requiring knowledge of the reliability function in advance. Compared with previous works to solve this problem, the proposed MCS-PSO can have better efficiency by providing a better solution to the complex network reliability optimization problem.**

***Index Terms*—Monte-Carlo simulation, network reliability, network reliability optimization, particle swarm optimization, series-parallel networks.**

ACRONYMS

MODA	multi-objective decision analysis
MC/MP	minimal cut/minimal path
MCS	Monte Carlo simulation
ANN	artificial neural network
PSO	particle swarm optimization
MCS-PSO	MCS based PSO
SqRSM	squeeze response surface methodology

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NOTATION

$G(V,E)$	a network with the node set V, and the arc set E; where nodes s, and t denote the source node, and sink node, respectively.
$\mathbf{R}$	the reliability vector, e.g. $\mathbf{r} = (r_1, r_2, \ldots, r_n)$ where r_i is the reliability of the ith component.
$P(\bullet)$, $E[\bullet]$, $Var[\bullet]$	the probability, expected value, and variance of $\bullet$, respectively.
R^{c}	the constraint of the lower bound of the system reliability.
r_i^{c}	the constraint of the lower bound of component r_i.
$C(\mathbf{r})$	the cost value under $\mathbf{r}$.
$R(\mathbf{r})$	the exact reliability value under $\mathbf{r}$.
$R^*(\mathbf{r})$	the approximate reliability value obtained from MCS under $\mathbf{r}$.
m, w, c_1, c_2	The parameters of the number of population (m), inertia weight (w), cognition learning factor (c_1), and social learning factor (c_2) using in PSO, respectively.
λ	The amplification parameter in the penalty function.
$R_i(\mathbf{r})$	the expected system reliability when removing i distinct nodes arbitrary (i distinct nodes are failed) under $\mathbf{r}$.
M	the number of simulation replications.

NOMENCLATURE

Reliability	is the probability that the system will operate successfully under the stated conditions for a stated duration of time.
MC/MP	is a cutset/path between nodes s and t such that any proper subset of cutset/path is not a cutset/path; i.e. if any arc is removed from the set, then the remaining set is no longer a cutset/path.

ASSUMPTIONS

1. Each component (arc or node) has two mutually exclusive states: working, and failed.
2. The component states are s-independent.
3. The graph is connected, and free of self-loops.

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I. Introduction

THE STUDY of reliability has received considerable interest in many areas. Examples of these areas are software reliability [1], [2], reliability of computing systems [3]–[6], and network reliability [40]. In particular, the reliability of networks has gained great impetus from its clear applicability to complex systems, and project schedules. Each component in these networks refers to an object in a complex network, or an operation in a project schedule. Network reliability is one of the useful decision support measures in management science. It has been applied in many real-world systems such as oil/gas production systems [7], computer and communication systems [8], [9], power transmission and distribution systems [10], and transportation systems [11]. Therefore, system reliability plays an important role in our modern society.

Reliability optimization has been a popular area of research, and received significant attention during the past four decades [7]–[11] due to the critical importance of reliability in various kinds of systems. Most of these works either assumed that the reliability functions are known in advance, or estimate the approximate symbolic network reliability function.

The network reliability optimization problem trades between cost and reliability. It can be simplified into two types of models without setting them as Multi-Objective Decision Analysis (MODA) problems. The first type sets the system reliability as the only goal, to maximize the system reliability with components' cost limitations; the second type is the dual problem, where the total cost is minimized under the system, and components' reliability constraints.

The network reliability problem is NP-hard, with computational effort growing exponentially with the number of nodes and links in the network. To achieve a better solution quality (computational efficiency, and estimation accuracy), modern meta-heuristics have been presented to solve complex network reliability optimization problems such as Response Surface Methodology [12], Artificial Neural Network (ANN) [13], Genetic Algorithms [14], [15], Ant Colony Optimization [16], [17], Tabu Search [18], [19], and Particle Swarm Optimization (PSO) [20], [21]. It is impractical, and very difficult to evaluate complex network reliability, and reduce the cost of the system at the same time. Hence, there is a need to develop a new method to solve the complex network reliability optimization problem such that the cost is minimized under the reliability constraints.

To solve reliability optimization problems, the reliability function is a necessary component under the above mentioned traditional methods. Computing the exact reliability of a network is also NP-hard [22]; therefore, estimation by simulation and other approaches often becomes an alternative choice. Monte-Carlo Simulation (MCS) is a straightforward simulation method for complex systems, and is now recognized as playing an important role in networks [23]. In addition, some works put forth other alternatives to the estimation of network reliability, such as response surface methodology [12], [24], and ANN [13].

MCS is one of the optimal algorithms to estimate the network reliability for different kinds of network configurations. Several authors have adopted MCS to measure the system reliability for conveniences, and accuracies. Kamat & Riley [25] introduced a general approach by combining the reliability of flow graph representation, Boolean state representation, and Monte Carlo simulation to estimate the system reliability. Their proposed methods require all Minimal Cuts (MCs) for the system in advance. Kubat [26] introduced a simulation/analytical approach to analyze communication and computer networks. Lin & Donaghey [27] presented a new MCS procedure for determining the Minimal Cut/Minimal Path (MC/MP), and system reliability. They use MCS to determine the MP by tracing through the system from input components to the output components, and then use MP to simulate the system failures. MC, and system reliability are determined by MCS approach. We have proposed a novel MCS without knowing MP/MC to estimate the network reliability directly in [24]. Furthermore, MCS methods have the following advantages when we use them to evaluate system reliability.

1. MCS methods can be applied to different kinds of network configuration such as series, parallel, and complex networks.
2. MCS can be applied to analyze systems whose components follow various distributions.
3. Faster computers are produced as technology progresses, so the manipulating time of MCS decreases [28]. As a result, the MCS algorithm has now become one of the more efficient, optimal approaches for estimating network reliability.

Most network reliability optimization problems are only focused on solving series-parallel networks (e.g., simple structured networks) of which the reliability function can be easy obtained in advance. PSO is a new population-based optimization algorithm. To our knowledge, the first published application of PSO in reliability engineering could be found in [20]. In this present paper, we have proposed to combine MCS with PSO to minimize the cost under reliability constraints. We have compared the proposed method with previous work for solving the related problems in [12], [24]. The proposed MCS-PSO method does not need to know the exact or approximate reliability functions in advance. This MCS-PSO algorithm only applies MCS to estimate the reliability for the specific particles when the fitness function is needed in PSO. Hence, the algorithm works better than traditional algorithms. In addition, the testing results show that the proposed method can have better efficiency in solving the complex network reliability optimization problem.

This paper is organized as follows. Section II describes the reliability optimization of which the reliability value can be easily obtained. Section III describes a MCS method without knowing MP/MC to estimate network reliability. Section IV introduces the traditional PSO, and a heuristic method. Section V proposes a new MCS-PSO which combines MCS with PSO to minimize the cost under reliability constraints. A numerical benchmark example for the network reliability is used in Section VI to demonstrate the MCS-PSO method. Section VII concludes with remarks.

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II. Traditional Reliability Optimization Problem

The traditional reliability optimization design is to determine a system structure which has the highest reliability for the minimum cost of manufacturing. This problem can be resolved using Yeh's paper [12], which is formulated as follows.

$$\text{Minimize} \quad C(\mathbf{r}) \tag{1}$$

$$\text{s.t.} \quad R(\mathbf{r}) \geq R^{\mathrm{c}} \tag{2}$$

$$\mathbf{r} = (r_1, r_2, \ldots, r_n) \geq \mathbf{r}^{\mathrm{c}} = (r_1^{\mathrm{c}}, r_2^{\mathrm{c}}, \ldots, r_n^{\mathrm{c}}) \tag{3}$$

The objective function is to minimize the cost. The first, and second constraint must satisfy the constraint of reliability. The reliability function of a system constructed by only simple series and parallel components can be obtained easily, but it cannot apply to most real-life networks as those networks are too complex. If it is a simple system constructed with only the series and/or parallel systems, then it can be determined by through combinations of the following four common system configurations [29].

Case 1) The simple series system:

$$R(\mathbf{r}) = \prod_{i=1}^{n} r_i. \tag{4}$$

Case 2) The simple parallel system:

$$R(\mathbf{r}) = 1 - \prod_{j=1}^{n} (1 - r_j). \tag{5}$$

Case 3) The parallel-series system:

$$R(\mathbf{r}) == 1 - \prod_{i=1}^{n} \left[1 - \prod_{j=1}^{n_i} r_{ij} \right]. \tag{6}$$

Case 4) The series-parallel system:

$$R(\mathbf{r}) = \prod_{i=1}^{n} \left[1 - \prod_{j=1}^{n_i} (1 - r_{ij}) \right]. \tag{7}$$

From above, most of the network reliability optimization problems are only focused on solving simple structured networks for which the reliability function can be easy obtained in advance.

III. MCS

MCS methods have been used to solve many network reliability problems. Many people have developed different kinds of Monte Carlo techniques to estimate the system reliability [25], [30]. Researchers have only applied MCS to estimate network reliability by focusing on how to design sampling plans to reduce the variance in terms of the known MP/MC [26], [31], [32]. However, to get all MP or MC information from the network is NP-Hard [33], [34]. Therefore, we have introduced a MCS procedure for a complicated system without the task of knowing MP/MC in [24], and we leverage that work in the present paper. It is worth noting that the MCS algorithm proposed in this paper is applicable to AOA (activity on arc), or/and AON (activity on node) network problems.

INPUT: A binary state network $G(V, E)$ with all nodes perfectly reliable, the source node s, the sink node t, the number of simulation replications M, and a given reliability vector $\mathbf{r}$.

OUTPUT: the estimator $R^*(\mathbf{r})$.

STEP 0. Set the current number of replications k, and the current number of successful simulation SUCCESS all equal to zero.

STEP 1. Let $S_0 = \phi$, and $S = \{1\}$.

STEP 2. Randomly choose a node in S, say i, and let $S^* = S \cup S_0$.

STEP 3. Randomly choose a node not in S^*, say j, such that $e_{ij} \in E$. If no such node exists, then go to STEP 7.

STEP 4. If arc e_{ij} is reliable (i.e., $r_{ij} = 1$), or $r_{ij} > r^*$, then go to STEP 5, where r^* is a random number generated from uniform(0,1). Otherwise, go to STEP 6.

STEP 5. If node j is reliable (i.e., $r_{ij} = 1$), or $r_{ij} > r^*$, then go to STEP 8, where r^* is a random number generated from uniform(0,1).

STEP 6. Let $S^* = S^* \cup \{j\}$, and go to STEP 3.

STEP 7. If $|S| = 1$, then go to STEP 9; else, let $S = S - \{i\}$, $S = S \cup \{i\}$, and go to STEP 2.

STEP 8. If $j = t$, then nodes s, and t are connected; let $\mathrm{SUCCESS} = \mathrm{SUCCESS} + 1$. Otherwise, let $S = S \cup \{j\}$, and go to STEP 2.

STEP 9. If $k < M$, then let $k = k + 1$, and go to STEP 1. Otherwise, let $R^*(\mathbf{r}) = \mathrm{SUCCESS}/M$, and halt.

The important statistical characteristics of $R^*(\mathbf{r})$ are summarized in the following properties. Their proofs can be found in many textbooks on reliability [31], [34].

Property 1: The expectation of the estimated reliability value obtained from MCS is an unbiased estimator for the exact reliability R.

Proof: Let $R_j^*(\mathbf{r})$ be the expected reliability of $G(V, E)$, where $j = 1, 2, \ldots, M$.

Then,

$$E\left[R_j^*(\mathbf{r})\right] = 1 \times P(\text{nodes } s \text{ and } t \text{ are connected}) + 0 \times P(\text{nodes } s \text{ and } t \text{ are disconnected}) = R(\mathbf{r}). \tag{8}$$

Thus, we have the estimator $R^*(\mathbf{r})$ of the system reliability

$$E\left[\sum_{j=1}^{m} \frac{R_j^*(\mathbf{r})}{m}\right] = \sum_{j=1}^{m} \frac{E\left[R_j^*(\mathbf{r})\right]}{m} = \sum_{j=1}^{m} \frac{R(\mathbf{r})}{m} = R(\mathbf{r}). \tag{9}$$

Property 2: $R^*(\mathbf{r})$ is an unbiased, consistent estimator of $R(\mathbf{r})$.

$$Var\left[R^*(\mathbf{r})\right] = \sum_{j=1}^{M} \frac{R(\mathbf{r})\left[1 - R(\mathbf{r})\right]}{M}. \tag{10}$$

Property 3: If the relative error ε, and the confidence interval $(1 - \alpha)\%$ for the simulation are required, then the total number

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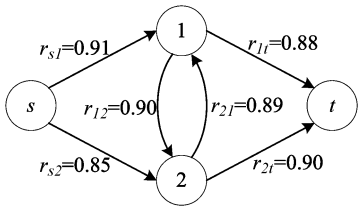

Fig. 1. The illustration network.

of replications of the simulation must be taken to be at least $M \geq (Z^2_{\alpha/2}/4\varepsilon^2)$.

The proposed MCS algorithm is illustrated using the following network problem shown in Fig. 1, under the assumption that each node is perfectly reliable.

STEP 0. Let $k = 0$, and $\mathrm{SUCCESS} = 0$.
STEP 1. Let $S_0 = \varnothing$, and $S = \{s\}$.
STEP 2. Randomly choose a node in S, say $i = s$, and let $S^* = S \cup S_0 = \{s\}$.
STEP 3. Randomly choose a node not in S^*, say $j = 1$, such that $e_{s1} \in E$.
STEP 4. Generate a random number from uniform(0,1), say $r^* = 0.5436$, and arc e_{s1} is reliable because $r_{s1} = 0.91 \geq r^* = 0.5436$.
STEP 5. Go to STEP 8 because all nodes are perfect in assumption.
STEP 8. Because $j = 1 \neq t$, let $S = S \cup \{1\} = \{s, 1\}$, and go to STEP 2.
STEP 2. Randomly choose a node in S, say $i = 1$, and let $S^* = S \cup S_0 = \{s, 1\}$.
STEP 3. Randomly choose a node not in S^*, say $j = t$, such that $e_{1t} \in E$.
STEP 4. Generate a random number from uniform(0,1), say $r^* = 0.9012$, and arc e_{1t} is not reliable because $r_{1t} = 0.88 \geq r^* = 0.9012$. And go to STEP 6.
SETP 6. Let $S^* = S^* \cup \{t\} = \{s, 1, t\}$, and go to STEP 3.
STEP 3. Randomly choose a node not in S^*, say $j = 2$, such that $e_{12} \in E$.
STEP 4. Generate a random number from uniform(0,1), say $r^* = 0.1424$, and arc e_{12} is reliable because $r_{12} = 0.90 \geq r^* = 0$.
STEP 5. Go to STEP 8 because all nodes are perfect in assumption.
STEP 8. Because $j = 2 \neq t$, let $S = S \cup \{2\} = \{s, 1, 2\}$, and go to STEP 2.
STEP 2. Randomly choose a node in S, say $i = 2$, and let $S^* = S \cup S_0 = \{s, 1, 2\}$.
STEP 3. Randomly choose a node not in S^*, say $j = t$, such that $e_{2t} \in E$.
STEP 4. Generate a random number from uniform(0,1), say $r^* = 0.3315$, and arc e_{2t} is reliable because $r_{2t} = 0.90 \geq r^* = 0.3315$.
STEP 5. Go to STEP 8 because all nodes are perfect in assumption.
STEP 8. Because $j = t$, let $\mathrm{SUCCESS} = \mathrm{SUCCESS} + 1$.
STEP 9. Next replication.

IV. Introduction to PSO

PSO is a new population based optimization algorithm, which was first introduced by Kennedy & Eberhart [35]. The goal of this algorithm is to optimize various continuous nonlinear functions. The concept of PSO is based on the metaphor of social interaction, and communication such as fish schooling and bird flocking.

In PSO, a solution is encoded as a finite-length string called "*a particle*". All of the particles have fitness values which are evaluated by the fitness function to be optimized, and have velocities which direct the flying of the particles. PSO is initialized with a population of random particles, with random positions and velocities inside the problem space, and then searches for optima by updating generations. It combines local search with global search, yielding high search efficiency. Each particle moves towards its best previous position, and towards the best particle in the whole swarm in every iteration. The former is a local best, and its value is called *pbest*; the latter is a global best, and its value is called *gbest*. After finding the two best values, the particle will update its velocity, and position with the equations

$$v_{ij}^t = w \cdot v_{ij}^{t-1} + c_1 \cdot \rho_1 \cdot \left(p_{ij}^{t-1} - x_{ij}^{t-1}\right) + c_2 \cdot \rho_2 \cdot \left(g_{ij}^{t-1} - x_{ij}^{t-1}\right), \tag{11}$$

and

$$x_{ij}^t = x_{ij}^{t-1} + v_{ij}^{t-1}. \tag{12}$$

The values $c_1\rho_1$, and $c_2\rho_2$ determine the weights of two parts, and the value of $(c_1 + c_2)$ is usually limited to 4 [32]. To apply PSO, several parameters including the population size (m), cognition learning factor (c_1), social learning factor (c_2), inertia weight (w), and the number of iterations or CPU time should be determined properly. We have conducted some preliminary experiments.

V. The Proposed MCS-PSO Method

This paper aims at creating a novel PSO by integrating the MCS. The proposed MCS-PSO can overcome the drawbacks of conventional methods which require finding all the exact or estimated reliability functions in advance to solve the complex network reliability optimization problem. The details of our proposed MCS-PSO method are presented below.

A. The Proposed Simple Heuristic Method to Find an Initial Solution

According to the model introduced in Section I, there are two constraints that must be satisfied. To satisfy the second constraint of the component reliability, we need to apply MCS to get the corresponding system reliability after deciding the components' reliability. However, this approach will add a penalty to the objective function. For example, it will add cost if the system reliability does not satisfy the lower bound of the system reliability constraint. The addition of cost will affect the solution we find. In this paper, a heuristic method to find the initial solution of a particle has been proposed, and the initial solution of the other remaining particles are still generated randomly. The main idea of this method is to find a feasible initial solution that

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satisfies the first and second constraints at the same time, and try to keep the components' reliabilities as small as possible. We prove in the next section that the proposed new initialization method is better than the traditional method in which all particles are generated randomly. The following will explain the detail of our proposed MCS-PSO method.

STEP 1. Assign the lower bound of the component reliability to the first particle.

STEP 2. Apply MCS to decide the corresponding system reliability $R^*(\mathbf{r})$.

STEP 3. If $R^*(\mathbf{r}) < R^{\mathrm{c}}$, then go to STEP 4. Otherwise, the corresponding components' reliabilities are the initial solution of this particle, so stop.

STEP 4. For each dimension of this particle:

STEP 4.1. Let the value of dimension i be $0.25^*(1 - r_i) + r_i$.

STEP 4.2. Apply MCS to get $R_i^*(\mathbf{r})$.

STEP 4.3. Adjust dimension i to its original reliability.

STEP 5. For each dimension of this particle:

STEP 5.1. If each dimension of $R_i^*(\mathbf{r})$ is less that the corresponding coordinate of R^{c}, then find the dimension with maximal value in $R_i^*(\mathbf{r})$, say j, and go to STEP 5.3.

STEP 5.2. Find the minimum $C(\mathbf{r})$ of dimension j such that $R_i^*(\mathbf{r}) \geq R^{\mathrm{c}}$, and go to STEP 5.3.

STEP 5.3. Let the value of dimension i be $0.25^*(1 - r_j) + r_j$, $R_j^*(\mathbf{r}) = R^*(\mathbf{r})$, and go to STEP 3.

B. Penalty Function

One of the major drawbacks of the MCS-PSO is that it may cause the system to have an infeasible structure, which will lead the algorithm to converge very slowly. This drawback implies that the simulation result of the system reliability $R^*(\mathbf{r})$ may not satisfy the lower bound of R^{c}. To prevent this drawback, and to guide the search towards the unexplored regions in the solution space, the penalty function has been implemented in the MCS-PSO. For a particle i with the total system cost C_i that does not satisfy R^{c}, a penalized cost is calculated as

$$C_i(r) \times \left(\frac{R^c}{R^*(r)}\right)^{\lambda} \text{ when } R^*(\mathbf{r}) < R^{\mathrm{c}} \tag{13}$$

where the exponent λ is an amplification parameter. This penalty function in (13) encourages the particles to explore both of the feasible region, and the infeasible region that is near the border of the feasible area, to help the search not to go too far into the infeasible region. This approach assures that the feasible, and infeasible regions of the search space are explored efficiently, and effectively to identify an optimal, or nearly optimal solution.

C. The Proposed MCS-PSO

In this paper, we propose a new MCS-PSO by using the MCS method which has been described in Section IV, and the PSO algorithm which has been presented in Section V, to solve the cost function problem under a reliability constraint. The explanations of the proposed MCS-PSO method are as follows.

STEP 0. Let $\mathrm{Iteration_Number} = 1$.

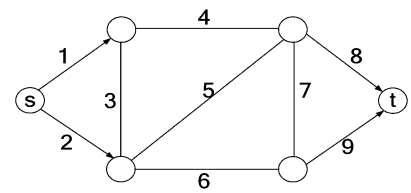

Fig. 2. The modified ARPA network.

TABLE I
THE COST FUNCTION OF EACH COMPONENT

Component i	α_i	β_i	$C(r_i)= \alpha_i - \beta_i \times \ln(1-r_i)$
1	120	14.7	$120 - 14.7\times\ln(1-r_1)$
2	120	14.7	$120 - 14.7\times\ln(1-r_2)$
3	90	8.75	$90 - 8.75\times\ln(1-r_3)$
4	100	9.9	$100 - 9.9\times\ln(1-r_4)$
5	65	5.64	$65 - 5.64\times\ln(1-r_5)$
6	100	9.9	$100 - 9.9\times\ln(1-r_6)$
7	90	8.75	$90 - 8.75\times\ln(1-r_7)$
8	160	12	$160 - 12\times\ln(1-r_8)$
9	160	12	$160 - 12\times\ln(1-r_9)$

TABLE II
THE SETTING OF MCS-PSO

Number of particle	15
Number of dimension	9
Max Velocity	1
Max Position	1
Iteration	100
cognitive factor	0.8
Social factor	0.8
Replication	30
Simulation replication	1000
λ	10

STEP 1. The first particle is initialized by heuristic method, and the other remaining particles are initialized randomly.

STEP 2. Apply MCS to find the corresponding $R^*(\mathbf{r})$ to each particle.

STEP 3. If $R^*(\mathbf{r}) < R^{\mathrm{c}}$, the cost will be determined by specific corresponding component reliability, and penalty function; otherwise, the cost will be determined by specific corresponding component reliability only.

STEP 4. Update the pbest, gbest, velocity, and position of each particle based on (11), and (12); and let $\mathrm{Iteration_Number} = \mathrm{Iteration_Number} + 1$.

STEP 5. If the stopping condition is true, that the maximum Iteration_Number is reached, then stop; else go to Step 2.

VI. EXPERIMENTAL RESULTS, AND SUMMARY

In this section, we use the proposed MCS-PSO to solve the cost minimization problem under reliability constraints. A numerical experiment has been conducted in the small-scale system. This problem is taken from Yeh [12], [24]. Fig. 2

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TABLE III
SOLUTIONS FOR THE PROPOSED MCS-PSO WITH DIFFERENT R^c AND $r^c = 0.8$

	R^c=0.9		R^c=0.95		R^c=0.99		R^c=0.925		R^c=0.975	
Replication	initialized randomly	heuristic method	initialized randomly	heuristic method	initialized randomly	heuristic method	initialized randomly	heuristic method	initialized randomly	heuristic method
1	1200.60	1160.05	1200.44	1183.10	1252.66	1234.90	1199.62	1171.19	1211.10	1209.85
2	1198.26	1160.05	1199.75	1186.18	1242.84	1236.61	1187.10	1171.19	1219.18	1208.35
3	1197.84	1160.05	1200.95	1182.90	1242.80	1234.01	1196.02	1171.19	1210.89	1208.08
4	1204.50	1164.28	1208.34	1182.84	1251.44	1235.08	1202.96	1171.85	1216.26	1208.62
5	1193.39	1160.05	1196.18	1182.71	1238.35	1238.35	1201.15	1171.96	1217.18	1206.11
6	1195.26	1160.05	1204.32	1183.09	1240.90	1237.28	1214.92	1171.19	1215.92	1207.08
7	1204.61	1160.05	1214.43	1182.02	1239.16	1239.73	1201.97	1171.96	1210.98	1207.28
8	1199.96	1160.05	1201.89	1181.87	1246.87	1237.16	1188.47	1171.19	1212.12	1207.97
9	1187.90	1160.05	1200.28	1183.10	1245.40	1239.02	1197.56	1171.19	1212.65	1206.76
10	1189.63	1160.05	1202.74	1182.68	1241.51	1235.58	1204.24	1171.16	1216.57	1206.82
11	1198.45	1164.28	1197.71	1182.32	1240.26	1237.68	1198.39	1170.32	1211.54	1210.13
12	1185.57	1160.05	1192.40	1183.10	1248.71	1237.50	1201.53	1171.96	1216.78	1206.02
13	1198.03	1160.05	1200.06	1186.67	1244.27	1236.49	1205.23	1171.96	1219.92	1206.37
14	1205.01	1160.05	1196.19	1183.10	1241.57	1237.93	1193.77	1171.75	1212.03	1206.96
15	1195.90	1163.51	1193.32	1182.50	1243.60	1236.10	1205.26	1171.96	1218.79	1207.83
16	1204.92	1164.28	1198.28	1183.10	1245.34	1235.61	1191.93	1171.19	1217.82	1206.95
17	1203.00	1163.51	1199.30	1187.32	1245.06	1236.86	1201.53	1171.41	1213.50	1207.31
18	1202.81	1163.51	1198.79	1185.27	1243.49	1235.85	1188.80	1171.19	1214.31	1205.40
19	1203.39	1164.28	1197.01	1185.31	1237.63	1233.74	1198.76	1171.19	1217.06	1207.68
20	1198.72	1164.28	1203.27	1183.09	1249.58	1236.57	1203.03	1171.96	1223.17	1207.31
21	1190.56	1160.05	1194.28	1182.91	1253.77	1239.22	1195.52	1171.19	1216.91	1207.63
22	1202.63	1163.51	1196.88	1182.23	1241.25	1234.71	1195.71	1171.19	1215.23	1207.86
23	1201.18	1164.28	1205.94	1182.14	1245.42	1237.37	1196.51	1170.50	1217.15	1206.08
24	1209.13	1160.05	1201.83	1185.95	1245.05	1238.01	1192.26	1171.96	1215.17	1204.62
25	1196.44	1164.28	1205.90	1182.48	1249.89	1235.24	1190.87	1171.96	1213.59	1206.96
26	1201.30	1160.05	1204.18	1182.86	1243.33	1235.08	1187.52	1171.96	1216.13	1206.93
27	1199.61	1160.05	1205.91	1185.88	1244.66	1235.68	1193.03	1171.96	1214.65	1206.80
28	1196.89	1163.51	1200.42	1186.55	1243.66	1234.57	1196.97	1171.19	1215.01	1206.01
29	1203.37	1163.51	1193.26	1182.25	1266.32	1236.58	1191.87	1171.96	1212.22	1204.91
30	1197.88	1160.05	1200.64	1182.55	1238.20	1239.74	1192.65	1171.19	1219.58	1205.31
Average	1198.89	1161.73	1200.50	1183.60	1245.10	1236.61	1197.17	1171.47	1215.45	1207.07
Varance	29.796	3.873	23.351	2.658	33.421	2.698	40.190	0.218	9.235	1.604
AvgError	3.348%	0.145%	1.564%	0.135%	0.846%	0.159%	2.430%	0.231%	0.885%	0.189%
ExactSol.	1160.05	1160.05	1182.01	1182.01	1234.65	1234.65	1168.77	1168.77	1204.79	1204.79
BestSol.	1185.57	1160.05	1192.40	1182.02	1237.63	1234.71	1187.10	1170.32	1210.89	1204.91
Error (%)	2.200%	0.000%	0.879%	0.001%	0.241%	0.005%	1.568%	0.132%	0.506%	0.010%

Note that $\mathrm{Error}\ (\%) = ((\mathrm{BestSol.} - \mathrm{ExactSol.})/\mathrm{ExactSol.}) \times 100\%$.

shows the benchmark example network, called the modified ARPANET. It is the most frequently cited example [36]–[38]. This network contains 6 perfectly reliable nodes, and 9 unreliable arcs. Table I has listed the cost function of each component.

The proposed MCS-PSO has been implemented in Visual Basic. The corresponding numerical parameters and data are presented in Table II for evaluating the performance, and confirming the validity of MCS-PSO. We use 15 particles, and each particle with 9 dimensions. The reliability is in the interval [0,1], so the maximum velocity, and the maximum position are both set to 1. We set the cognitive factor, and the social factor both to 0.8 because the reliability is within [0,1]. In the testing experiments, we repeated each test 30 times. We apply the MCS method for each particle with 1000 replications to get $R^*(\mathbf{r})$. The penalty function will be applied only if the simulation result $R^*(\mathbf{r})$ does not satisfy the lower bound of the constraint R^c. We use the penalty function [16]

$$C(\mathbf{r}) \times \left(\frac{R^c}{R^*(\mathbf{r})}\right)^{\lambda} \tag{14}$$

where $C(\mathbf{r})$ is the corresponding cost under specific components' reliabilities, and λ is a constant value. By trial, we set λ to 10 in this problem.

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TABLE IV
SOLUTIONS TO THE COST MINIMIZATION BENCHMARK PROBLEM WITH $R^c = 0.9$

	R^c=0.9, r_i^c=0.8			
Result	ANN	SqRSM	MCS-PSO	Exact
r_1	0.80000	0.80000	0.80000	0.80000
r_2	0.80000	0.80000	0.80000	0.80000
r_3	0.80000	0.80000	0.80000	0.80000
r_4	0.80000	0.80000	0.80000	0.80000
r_5	0.97639	0.80000	0.80000	0.80000
r_6	0.80000	0.80000	0.80000	0.80000
r_7	0.80000	0.80000	0.80000	0.80000
r_8	0.80000	0.80000	0.80000	0.80000
r_9	0.80000	0.80000	0.80000	0.80000
$R^m(\mathbf{r})$	0.90889	0.90347	0.90000	0.90347
$C(\mathbf{r})$	1172.10	1160.05	1160.05	1160.05

TABLE V
SOLUTIONS TO THE COST MINIMIZATION BENCHMARK PROBLEM WITH $R^c = 0.95$

	R^c=0.95, r_i^c=0.8			
Result	ANN	SqRSM	MCS-PSO	Exact
r_1	0.80000	0.80000	0.80245	0.80000
r_2	0.87592	0.89418	0.90043	0.90072
r_3	0.80000	0.80000	0.80200	0.80000
r_4	0.80000	0.80000	0.80587	0.80000
r_5	0.91200	0.80000	0.80547	0.82838
r_6	0.86073	0.80000	0.80530	0.80000
r_7	0.80000	0.80000	0.80172	0.80000
r_8	0.82388	0.91706	0.91562	0.91864
r_9	0.93383	0.80000	0.80483	0.80000
$R^m(\mathbf{r})$	0.95378	0.95000	0.95000	0.95276
$C(\mathbf{r})$	1190.08	1180.00	1182.02	1182.01

TABLE VI
SOLUTIONS TO THE COST MINIMIZATION BENCHMARK PROBLEM WITH $R^c = 0.99$

	R^c=0.99, r_i^c=0.8			
Result	ANN	SqRSM	MCS-PSO	Exact
r_1	0.96804	0.80000	0.90554	0.80000
r_2	0.94358	0.97984	0.96114	0.98036
r_3	0.85709	0.80000	0.80919	0.80000
r_4	0.84001	0.84024	0.89422	0.80000
r_5	0.97254	0.94360	0.84053	0.96722
r_6	0.98338	0.80000	0.88214	0.80000
r_7	0.82879	0.80000	0.81047	0.80000
r_8	0.90274	0.98448	0.94388	0.98395
r_9	0.93795	0.80000	0.91728	0.80000
$R^m(\mathbf{r})$	0.99062	0.99000	0.99000	0.99086
$C(\mathbf{r})$	1270.65	1233.80	1234.71	1234.65

Table III shows the results of the proposed MCS-PSO. The randomly initialized particles can have different lower bound system reliability values. This heuristic method applied MCS with 5000 replications.

TABLE VII
SOLUTIONS TO THE COST MINIMIZATION BENCHMARK PROBLEM WITH $R^c = 0.925$

	R^c=0.925, r_i^c=0.8			
Result	ANN	SqRSM	MCS-PSO	Exact
r_1		0.80000	0.80244	0.80000
r_2		0.83045	0.85540	0.84237
r_3		0.80000	0.80125	0.80000
r_4		0.80000	0.80791	0.80000
r_5		0.80000	0.80523	0.80000
r_6		0.80000	0.80282	0.80000
r_7		0.80000	0.80124	0.80000
r_8		0.86699	0.86222	0.87052
r_9		0.80000	0.80074	0.80000
$R^m(\mathbf{r})$		0.92500	0.92500	0.92500
$C(\mathbf{r})$		1167.38	1170.32	1168.77

TABLE VIII
SOLUTIONS TO THE COST MINIMIZATION BENCHMARK PROBLEM WITH $R^c = 0.975$

	R^c=0.975, r_i^c=0.8			
Result	ANN	SqRSM	MCS-PSO	Exact
r_1		0.80000	0.81338	0.80000
r_2		0.94969	0.95168	0.95069
r_3		0.80000	0.80615	0.80000
r_4		0.80000	0.80385	0.80000
r_5		0.88551	0.81451	0.91663
r_6		0.80000	0.86234	0.80000
r_7		0.80000	0.80938	0.80000
r_8		0.96058	0.91101	0.95967
r_9		0.80000	0.89929	0.80000
$R^m(\mathbf{r})$		0.97500	0.97400	0.97500
$C(\mathbf{r})$		1202.98	1204.91	1204.79

From Table III, we find that the solutions using the heuristic method to find the initial solution of the first particle to perform better than the method that initializes all particles randomly. In addition, our proposed method can have lower variance, lower average error, and the best solution closest to the optimal solution. The optimal solution is referenced from Yeh [12]. The following are the hypothesis tests to determine whether the solutions using the heuristic method to find a particle's initial solution while the others remain randomly initialized will be better than the traditional way of having all particles initialized randomly. Let μ_{r} be an average solution of initializing randomly, and μ_{h} be an average solution of a particle initialized by a heuristic method. The test is then

$$\alpha = 0.05$$
$$\mathrm{H}_0 : \mu_{\mathrm{r}} \leq \mu_{\mathrm{h}}$$
$$\mathrm{H}_1 : \mu_{\mathrm{r}} > \mu_{\mathrm{h}}$$

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TABLE IX
COMPARISON OF ANN, SQRSM, AND THE PROPOSED MCS-PSO WITH DIFFERENT R^c

	R^c=0.9, r^c=0.8		R^c=0.925, r^c=0.8		R^c=0.95, r^c=0.8		R^c=0.975, r^c=0.8		R^c=0.99, r^c=0.8	
	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)
ANN	1172.100	1.038%			1190.080	0.683%			1270.650	2.916%
SqRSM	1160.053	0.000%	1167.376	-0.119%	1180.000	-0.170%	1202.976	-0.150%	1233.800	-0.069%
PSO(average)	1161.730	0.145%	1171.467	0.231%	1183.601	0.135%	1207.065	0.189%	1236.608	0.159%
PSO(the best)	1160.053	0.000%	1170.316	0.132%	1182.017	0.001%	1204.911	0.010%	1234.707	0.005%
Exact solution	1160.053		1168.771		1182.010		1204.787		1234.650	

TABLE X
THE STATISTICAL RESULTS FOR THE PROPOSED MCS-PSO

	R^c=0.9, r^c=0.8		R^c=0.925, r^c=0.8		R^c=0.95, r^c=0.8		R^c=0.975, r^c=0.8		R^c =0.99 r^c =0.8	
	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)	$C(\mathbf{r})$	Error(%)
Q1 of the first 10 data sets	1160.053	.000%	1171.186	.207%	1182.688	.057%	1206.882	.174%	1235.207	.045%
Q1 of the first 20 data sets	1160.053	.000%	1171.187	.207%	1182.700	.058%	1206.802	.167%	1235.604	.077%
Q1 of all 30 data sets	1160.053	.000%	1171.187	.207%	1182.516	.043%	1206.172	.115%	1235.320	.054%
the average of the first 10 data sets	1160.476	.036%	1171.406	.225%	1183.048	.088%	1207.692	.241%	1236.772	.172%
the average of the first 20 data sets	1161.628	.136%	1171.447	.229%	1183.612	.136%	1207.443	.220%	1236.603	.158%
the average of all data sets	1161.730	.145%	1171.467	.231%	1183.601	.135%	1207.065	.189%	1236.608	.159%
the best of the first 10 data sets	1160.053	.000%	1171.156	.204%	1182.017	.001%	1206.105	.109%	1235.084	.035%
the best of the first 20 data sets	1160.053	.000%	117.316	.132%	1182.017	.001%	1205.399	.051%	1235.084	.035%
the best of all data sets	1160.053	.000%	117.316	.132%	1182.017	.001%	1204.911	.010%	1234.707	.005%
Exact solution	1160.053		1168.771		1182.010		1204.787		1234.650	

Note that $\text{Error}\ (\%) = [((\text{Obtain solution} - \text{Exact solution})/\text{Exact solution})] \times 100\%$ in Tables IX, and X.

We use a Wilcoxon Rank Sum Test to test these two populations, as they do not fit the normal distribution.

$$\text{If } R^c = 0.9, \mathrm{W} = 1365.0, \quad \text{and P} - \text{value} < 0.05, \text{then reject H}_0.$$
$$\text{If } R^c = 0.95, \mathrm{W} = 1365.0, \quad \text{and P} - \text{value} < 0.05, \text{then reject H}_0.$$
$$\text{If } R^c = 0.99, \mathrm{W} = 1344.0, \quad \text{and P} - \text{value} < 0.05, \text{then reject H}_0.$$

From the above results, we can conclude that the solutions using the heuristic method to find one particle's initial solution while keeping others randomly initialized can perform better than all particles initialized randomly, plus it gives a smaller error, and a solution closer to the optimal, best solution.

The solutions to the cost minimization benchmark problem were shown in Tables IV–VIII with different R^c. The results of ANN, SqRSM, and exact solutions are referenced from previous work [12], [39]. If we want to calculate the solutions for ANN, SqRSM, and exact solutions, we must know the ANN approximate function, the SqRSM function, and the exact reliability structure functions in advance. Therefore, MATLAB or Lingo software is adopted to solve the above nonlinear programming model. However, the proposed MCS-PSO can obtain the solutions without knowing the approximate reliability function. The MCS-PSO solutions in Tables IV–VIII are the best solutions from Table III. The objective value of MCS-PSO is the closest solution to the exact value between these three methods. In Table IV, the reason for finding the optimal solution of MCS-PSO is that we have used a heuristic method to find the first particle's initial optimal solution instead of using a random number. In Table VII, the best solutions obtained from SqRSM, and MCS-PSO are 1167.38, and 1168.77, respectively. It seems that SqRSM outperforms MCS-PSO in this case. However, the best solution (i.e. cost) obtained from SqRSM is underestimated, and it may caused a fatal problem in real-life application. Hence, MCS-PSO is still better than SqRSM in Table VII. Similar situations also can be found in Table VIII. Table IX also shows the results of ANN, SqRSM, the average solution of MCS-PSO, and the best solution of MCS-PSO. By the best solution of MCS-PSO, we mean the closest to the exact solution. SqRSM is better than the average solution of MCS-PSO and ANN when R^c is equal to 0.9, and 0.99. When R^c is equal to 0.95, the average solution of MCS-PSO is better than SqRSM, and ANN.

Table X, and Fig. 3 show different kinds of data results with different R^c, and different amounts of data when applying MCS-PSO to find the minimum cost, and the corresponding component reliability. We make several observations from Fig. 3.

- SqRSM always underestimates the cost except $R^c = 0.9$ [see Fig. (3a)].
- There is no significant change when we increase the simulation number of MCS-PSO.
- The data of Q1 can be used to represent the overall result of MCS-PSO because the lower the cost we get, the closer to the exact solution. Therefore, we can use the first 25 percent of data to represent for the whole set of 30 data.
- The best solution of MCS-PSO is better than the best solution of SqRSM.
- The data of Q1 sometimes is better than SqRSM when $R^c = 0.95$, and $R^c = 0.99$ [see Figs. (3c) and (3e)].

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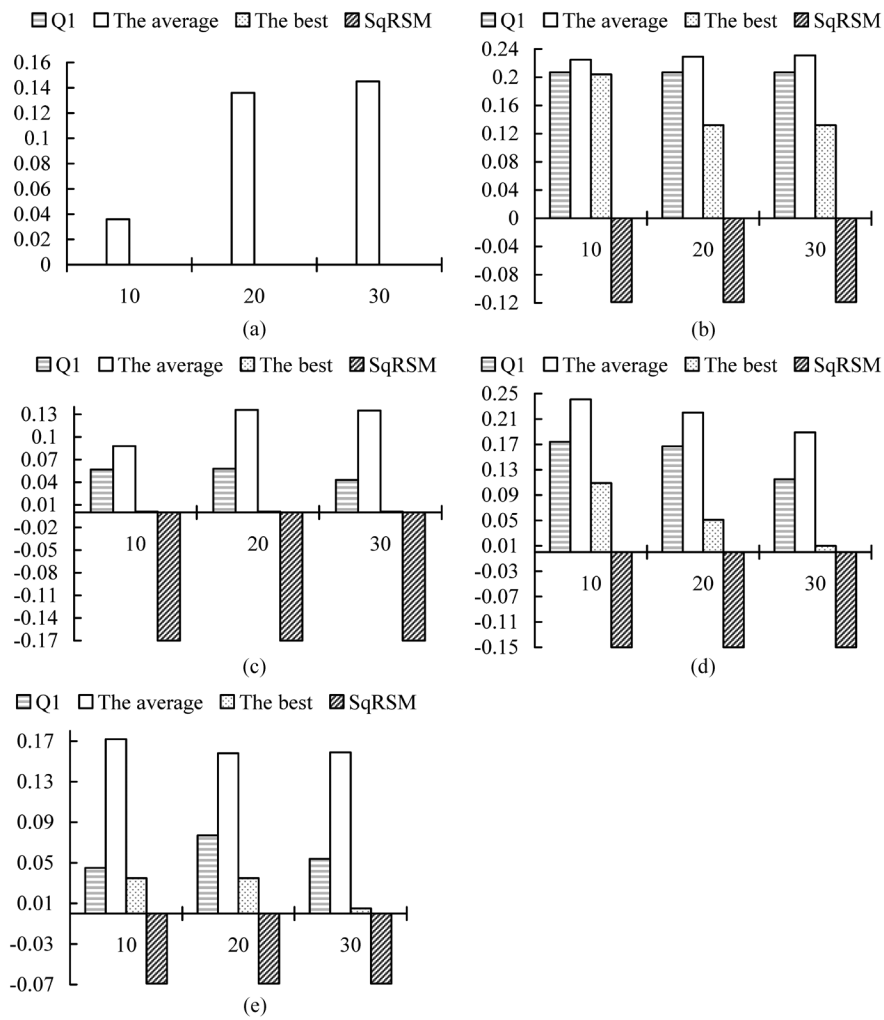

Fig. 3. The result of MCS-PSO, and SqRSM with different R^c, where the vertical axis denotes the error (%), and the horizontal axis denotes the data set. (a) $R^c = 0.9$. (b) $R^c = 0.925$.(c) $R^c = 0.95$. (d) $R^c = 0.975$. (e) $R^c = 0.99$.

VII. Conclusions, and Future Research

In this paper, a new MCS-PSO was proposed for solving for complex network reliability by minimizing the cost of components that constituted the network under reliability constraints. Compared with previous methods [12], [24], MCS-PSO does not require knowing the approximate reliability functions to solve this network problem. From experiment results, MCS-PSO has proved to have better efficiency in solving the complex network reliability optimization problem as it can provide a solution which is closer to the exact solution.

To our knowledge, this is the first reported study to combine MCS & PSO to solve complex network reliability. However, there are other issues that merit future research; for instance, the mathematic properties for MSC or PSO, such as variance reduction techniques, advanced sampling methods, and heuristic convergent properties. We hope the proposed MCS-PSO could also be extended to solve more complex reliability optimization problems, such as multi-state/commodity network reliability problems, in the near future.

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